

George Saad

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Dear Hiring Manager,

I am excited to apply for the Developer position, where I can leverage over five years of combined research, clinical and software development experience to support projects through their full lifecycle. My proficiency in Python, JavaScript, React.js, Flask, SQL and NO-SQL databases and over 2 years of cloud developments, paired with hands-on experience in clinical and academic research environments, uniquely positions me to collaborate with faculty to understand their needs and deliver robust, user-friendly applications for both mobile and web platforms.

In my current role at Dr. Ada Eban-Rothschild's Lab (AER Lab), I replicated and enhanced a spindle-detection convolutional neural network (CNN) model to predict sleep states using EEG and EMG recordings, achieving an in-lab 97.25% accuracy. To broaden accessibility, I designed and implemented a Python-based GUI that allows graduate students and researchers to run the model end-to-end without code knowledge or initial setups. This tool not only accelerated data analysis but also standardized our sleep-scoring workflow across the lab, reducing the manual workflow by 90%, from 4 hours to 23 minutes.

I am well-versed in Agile and Scrum methodologies, having participated in sprint planning, daily stand-ups, and retrospectives while working at LM Credit to ensure timely delivery of project milestones. At LM Credit, I routinely use Git for version control and Jira for task and story management and maintain comprehensive API and platform documentation to facilitate accessibility. Additionally, I have automated routine KPI reporting by leveraging Python web-scraping with Selenium, reducing manual effort by over 65% on daily, weekly, and monthly metrics.

Beyond software development, I have contributed to a variety of technical innovations in the lab. I assisted in designing and deploying an automated sound-dampened recording chamber using Raspberry Pi and RealVNC, enabling remote experiment monitoring. I supported the setup and upkeep of a FED3 pellet-feeding device via Arduino for behavioral assays. I also led a lab-wide data cleanup, deduplication, and backup initiative across the University of Michigan's Great Lakes storage systems—managing upwards of 100 TB of archived data. Furthermore, I developed and refined AccuSleep, a MATLAB-based sleep-scoring GUI that has become our lab's standard analysis tool, showcasing my ability to learn other languages and adapt to research needs.

I am drawn to your program's collaborative environment and commitment to innovation. I am confident that my blend of research insight, software craftsmanship, and operational rigor will help advance your mobile and web-based research applications. Thank you for considering my application. I look forward to the opportunity to discuss how my background and skills can contribute to your team's success.

Sincerely,

George

George Saad

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<http://github.com/gsaad>

Full-Stack Developer with 3+ years building production-grade web and mobile applications and 5+ years of psychology & neuroscience research experience. Skilled at translating complex user needs into intuitive interfaces and data-driven solutions. Proven track record of collaborating in cross-functional teams to deliver high-impact projects, from HIPAA-compliant data pipelines to responsive React applications.

Key Technical Skills

- **Front-End:** JavaScript, TypeScript, React, Angular, Next.js, HTML5, CSS3, Tailwind CSS
- **Back-End:** Node.js (Express, Fastify), Python (Flask, Django)
- **Mobile:** React native, Flutter (Dart)
- **Databases:** PostgreSQL, MongoDB, MySQL, Firebase
- **Cloud & DevOps:** AWS S3, Google Cloud Storage, Azure Blob Storage, Git, Jira, Heroku
- **Data Science & ML:** Pandas, NumPy, Matplotlib, PyTorch, LangChain, prompt engineering

EDUCATION

Full Stack Web Development Program
University of Toronto

Feb 2022 - Aug 2022
Toronto, ON

Bachelor of Science in Psychology & Neuroscience
Wilfrid Laurier University

Sep 2015 – Apr 2020
Waterloo, ON

EXPERIENCE

University Of Michigan - Research Technician Intermediate

Jun 2024-Present
Ann Arbor, MI

- Developed PowerShell and Python scripts utilizing SLURM and Great Lakes U-Mich clusters to parallelize code, scan and organize large data, tag files, and identify duplications for efficient storage management.
- Maintained standard operating procedures (SOPs) for lab compliance and efficiency.
- Implemented experiments for graduate students, following strict protocols in a timely manner.
- Managed and curated experimental data archives, ensuring organization and accessibility for auditing and reviewing research publications.
- Replicated a spindle detection Convolutional Neural Network (CNN) model and designed a GUI for sleep state predictions, achieving 97.25% accuracy.
- Experienced in histological techniques [perfusions, brain tissue slicing, mounting], imaging, and Adobe Photoshop enhancements for neuroscience research publications.
- Prepped EEG & EMG mounting, fiber photometry, and viral injection surgeries for experiments.
- Assisted in the design and construction of 3D-printed lab structures using Fusion360, infrared camera experimental setups, and novel experimental behavioral maze.
- Assisted in designing an automated SD soundproofed chamber using Raspberry Pi and RealVNC for remote operations.

- Assisted in the setup and maintenance of a FED3 feeding device utilizing Arduino.
- Led lab-data cleanup, deduplication, and backups across local and U-Mich Great Lakes storage systems, up to 100TB of data.
- Developed and enhanced AccuSleep tool, a MATLAB based scoring GUI as the lab standard tool.

LM Credit –Software Developer / IT (Tier 3)

**Apr 2023-Present
Toronto, ON**

- Developed role-based access to components using Angular.js Frontend, MSSQL database with a Node.js backend integration.
- Utilized Python for web scraping, selenium to automate (KPIs) D/W/M.
- Responsible for user onboarding, including account setup through a domain controller (DC) utilizing Active Directory (AD) and monitoring network and security audits.
- Attained Azure training which facilitated the migration of an on-premises server to cloud while setting up disaster recovery and daily backups of databases and virtual machines.
- Proficient in Git and Jira for task and story management, while maintaining a comprehensive API and platform documentation, facilitating team collaboration and knowledge sharing.
- Participated in SCRUM and AGILE techniques to enhance project management and collaboration.
- Collaborated with diverse departments and managements to align development efforts with strategic business objectives and stakeholder interests.

**EdX – Teaching Assistant-Full Stack
Present Toronto, ON**

Jul 2022-

- Collaborated closely with students in guiding their learning using a range of technologies like MongoDB, MySQL, Express.js, Node.js, and React while addressing student inquiries related to programming concepts and industry standards
- Led scrum-like environments to facilitate learning and fast paced production programming.
- Delivered daily and weekly communications with announcements and reports to students and colleagues, engaging the students and allowing them to organize their workload.
- Increased engagement and attention of students during class to facilitate understanding.

**Freelance – Full Stack Developer
Self-employed**

**Jun 2021-Present
Toronto, ON**

- Refined responsive layouts as clients specifications to ensure proper UI/UX for mobile phones, laptops and larger screens.
- Assisted in the production of an open-source project instance OscarEMR for a local business to facilitate invoices, faxes, and transactions between medical clinics and partnered pharmacies.
- Developing applications using React, Angular frontend with MongoDB and SQL backend utilizing Python and Node backend.
- Broadening knowledge in novel technologies and techniques to facilitate efficiency or monetization.
- Utilized Jira for task management and measuring code efficiency

Certifications

AWS Cloud Practitioner

AWS Developer Associate [Acquiring]

AWS Machine Learning Engineer Associate [Acquiring]